

Jesse Noppe-Brandon

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EDUCATION

New York University, Tandon School of Engineering, Brooklyn, NY

December 2025

Master of Science, Computer Science, GPA: 3.73

Relevant Coursework: Principles of Database Systems, Programming Languages, Data Science, Design and Analysis of Algorithms, Operating Systems, Machine Learning, Big Data, Software Engineering, Deep Learning, Cloud Computing, Responsible AI, AI in Education

- New York University Tandon Bridge Program

December 2023

CUNY Graduate Center (Interdisciplinary BA Program), New York, NY

August 2019

Bachelor of Arts, Major: Comparative Thought, Magna Cum Laude, Dean's List

TECHNICAL SKILLS

Coding Languages & Web Dev: Python, C++, SQL, Julia, HTML, CSS, Next.js, Flask, Django, FastAPI

Databases & Big Data: MySQL, Apache Spark, Hadoop, Kafka, MongoDB, PostgreSQL

Cloud & Tools: AWS, Git, Docker, Kubernetes, UNIX, Linux, Cursor, VSCode

AI & ML: OpenAI API, RAG, LangChain, LangGraph, PyTorch, Scikit-Learn

PROJECTS

InterviewAI (OpenAI, AWS: Lambda, API Gateway, Cognito, DynamoDB, RDS, CloudFormation, CodePipeline)

- Architected a serverless microservices system for an AI technical interviewer that recommends questions based on user history and generates personalized hints on the user's current code context using OpenAI API
- Coordinated with a team of four through GitHub and CloudFormation documents
- Spearheaded backend development by implementing AI Logic Lambdas, provisioning RDS instances, and optimizing SQL queries to connect to the frontend using API Gateways, Cognito and DynamoDB.
- Ensured availability through AWS CodePipeline and scalability through the AWS toolkit

ParkEasy (Django, AWS Elastic Beanstalk, Postgres, Travis CI, Zenhub, Coveralls)

- Built a parking spot sharing application using Django and Postgres, and deployed it on Elastic Beanstalk
- Participated in daily scrum meetings and weekly sprints in a team of 6, cycling the role of scrum master and managing our tasks through Zenhub
- Maintained 95% code coverage on Coveralls and 100% build availability with Travis CI

Pack (MongoDB Atlas, LangGraph, OpenAI, Next.js, FastAPI, Docker, AWS Lightsail, Vercel, S3, Clerk, GitHub Actions)

- Architected a multi-tenant family archive platform to digitize and query physical history, implementing a "Parent-Document" indexing strategy that utilizes GPT-4o Vision to transcribe handwriting while preserving full narrative context for the LLM.
- Engineered a stateful Agentic RAG system using LangGraph and MongoDB, capable of intelligently routing queries between deterministic metadata filtering and semantic vector search based on real-time intent classification.
- Designed a containerized hybrid-cloud infrastructure, deploying a Dockerized FastAPI backend to AWS Lightsail and a Next.js frontend to Vercel, utilizing GitHub Actions for fully automated CI/CD pipelines.
- Implemented strict multi-tenant security by injecting Organization IDs into every vector operation and built a responsive streaming interface using Clerk auth and AWS S3 presigned URLs for secure, temporary media access.

EXPERIENCE

Teaching Assistant, NYU Principles of Database Systems, New York, NY

September 2025 - December 2025

- Collaborated with faculty to update assignments, created official solutions, and developed rubrics
- Lead demo sessions on SQL, Flask, HTML, and CSS for 30 undergraduate students

Head Teaching Assistant, NYU Tandon Bridge, New York, NY

May 2024 - December 2025

- Managed a team of 7 Teaching Assistants, and lead in recruitment, onboarding, and training processes
- Oversaw administrative tasks, including configuring course applications and ensuring smooth onboarding for incoming students
- Hosted weekly online office hours to help 100+ students transition into STEM with subjects like discrete math, introductory C++, object-oriented programming, operating systems, and networking
- Debugged and evaluated C++ and Operating Systems implementations for 100+ students, identifying logic errors and memory leaks
- Collaborated with professors and other TAs to design and refresh materials using Powerpoint and quiz materials using Excel and Brightspace for an update to the course starting January 2026

Teacher, New York Language Center, New York, NY

February 2021 - August 2024

- Instructed classes of 10 to 50 adult international students daily across various levels, from beginner to TOEFL preparation; designed lesson plans and conducted assessments to evaluate student progress